Condensation and Metastability in the supercritical two-species zero-range process via resolvent and H^1-approximation

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Joint work with Seonwoo Kim

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SNU Probability seminar

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Contents

1. Classical zero-range process
2. Two-species zero-range process
3. Main theorem

Classical zero-range process

- **Random walk on finite space** S
  - ⋆ Irreducible jump rate $r : S \times S \to [0, \infty)$.
  - ⋆ Normalized stationary measure m, i.e, $\max_{x \in S} m(x) = 1$.
  - ⋆ $S_\star = \{x \in S : m(x) = 1\}$
- **Configuration space** $\Omega_N = \{\eta \in \mathbb{N}_0^S : |\eta| = N\}$
  - ⋆ $\eta = (\eta(x))_{x \in S}$
  - ⋆ $\eta(x) \in \{0, 1, \dots, N\}$
- **Dynamics** each particle jumps $x \to y$ at rate $g(\eta(x))r(x, y)$

$$g(n) = \begin{cases} \frac{\mathfrak{a}(n)}{\mathfrak{a}(n-1)} & \text{if } n \geq 1 \\ 0 & \text{if } n = 0 \end{cases}, \quad \mathfrak{a}(n) = \begin{cases} n^{\alpha} & \text{if } n \geq 1 \\ 1 & \text{if } n = 0 \end{cases}.$$

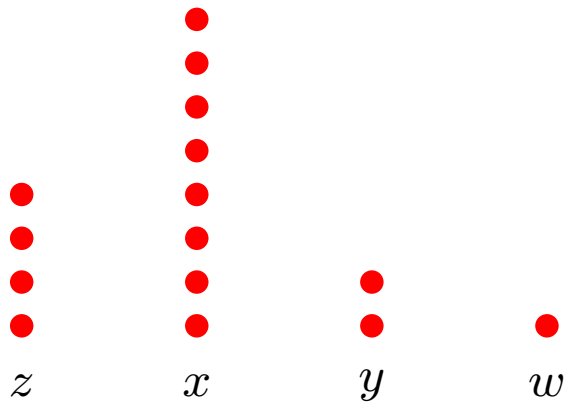
- **Stationary Distribution** $\mu_N(\eta) \propto \prod_{x \in S} \frac{m(x)^{\eta(x)}}{\mathfrak{a}(n)}, \quad \eta \in \Omega_N$

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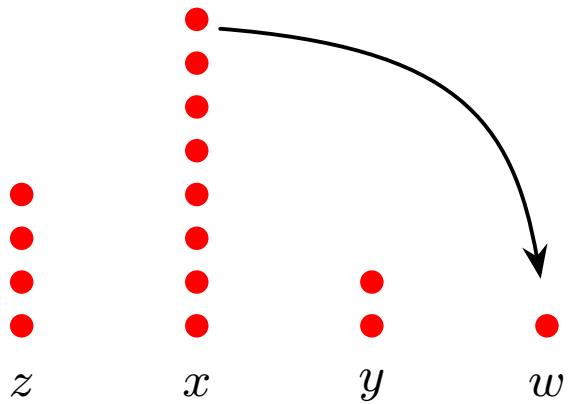


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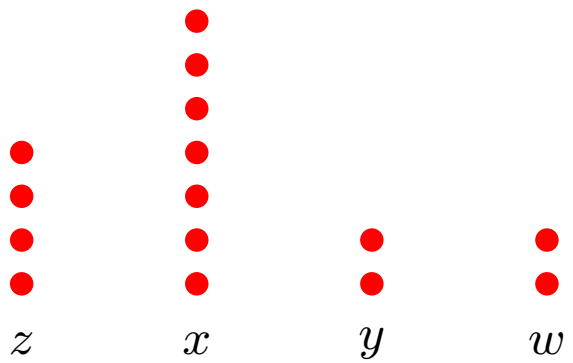


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Condensation

- **Regime**

α	Regime
$0 < \alpha < 1$	Sub-critical
$\alpha = 1$	Critical
$\alpha > 1$	Super-critical

- **Metastable well** For each $x \in S$ and $1 \ll \ell_N \ll N$,

$$\mathcal{E}_N^x := \{\eta : \eta(x) \geq N - \ell_N\}, \quad \mathcal{E}_N := \bigcup_{x \in S_\star} \mathcal{E}_N^x$$

- **Negligible set** $\Delta_N := \Omega_N \setminus \mathcal{E}_N$

Condensation

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Condensation (Beltrán–Landim '12, Landim–Marcondes–Seo '23)

⋆ For $\alpha \in (1, \infty)$,

$$\lim_{N \to \infty} \mu_N(\mathcal{E}_N^x) = \frac{1}{|S_\star|} \quad (x \in S_\star) \quad \text{and} \quad \lim_{N \to \infty} \mu_N(\Delta_N) = 0. \tag{1}$$

⋆ If $\alpha = 1$, condensation also occurs if ℓ_N additionally satisfies

$$\lim_{N \to \infty} \frac{\log \ell_N}{\log N} = 1.$$

⋆ If $\alpha \in (0, 1)$, there is no condensation phenomenon, i.e

$$\lim_{N \to \infty} \mu_N(\mathcal{E}_N^x) = 0, \quad x \in S$$

Markov chain reduction

- **Trace Process**

$$T_N(t) := \int_0^t \mathbb{1}\{\eta_N(s) \in \mathcal{E}_N\} ds, \quad S_N(t) := \sup\{s \geq 0 : T_N(s) \leq t\}, \quad t \geq 0.$$

Then *trace process* on $\mathcal{E}_N$ is defined as

$$\eta_N^{\mathrm{tr}}(t) := \eta_N(S_N(t)).$$

- **Order process** To track the massive condensate.

$$\Psi_N : \mathcal{E}_N \to S_\star, \quad \text{where} \quad \Psi_N = \sum_{x \in S_\star} x \mathbb{1}_{\mathcal{E}_N^x}.$$

Then *order process* on $S_\star$ is defined by

$$X_N(t) = \Psi_N(\eta_N^{\mathrm{tr}}(t)).$$

Effective transition

Regime	**Time scale θ_N**	**Limiting Markov chain $\mathbb{X}$ on $S_\star$**
$\alpha > 1$	$N^{\alpha+1}$	$\dfrac{1}{\Gamma(\alpha)\mathcal{I}_\alpha}\,\mathrm{cap}(x,y)$
$\alpha = 1$	$N^2 \log N$	$6\,\mathrm{cap}(x,y)$

Theorem (Metastability)

1. ***Local metastability*** $\lim_{N\to\infty} \inf_{\eta\in\mathcal{E}_N^x} \mathbb{P}_\eta^N[H_{\xi_N^x} < H_{\mathcal{E}_N\setminus\mathcal{E}_N^x}] = 1.$
2. ***Convergence*** $\{X_N(\theta_N t)\}_{t\geq 0} \xrightarrow[N\to\infty]{} \{\mathbb{X}(t)\}_{t\geq 0}.$
3. ***Negligibility***

$$\lim_{N\to\infty} \sup_{\eta\in\mathcal{E}_N^x} \mathbb{E}_\eta^N \left[\int_0^T \mathbb{1}\{\eta_N(\theta_N t)\in\Delta_N\}dt\right] = 0, \quad \forall\, T > 0.$$

Effective transition

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Known results

	Reversible	**Non-reversible**
$\alpha > 1$	Beltrán–Landim (2012)	Seo (2019)
$\alpha = 1$	Landim–Marcondes–Seo (2025)	**Open**

Contents

1 Classical zero-range process

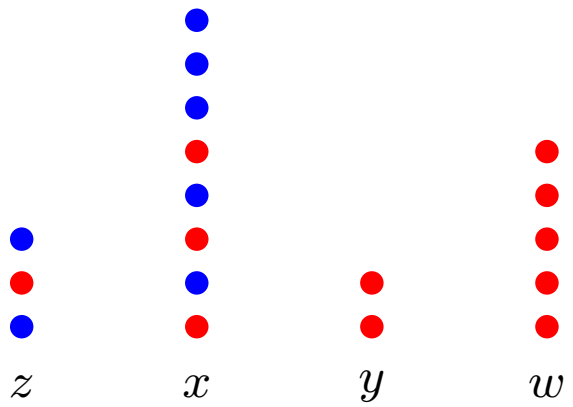
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Two-species zero-range process

- **Two random walks on finite space S**
  - ⋆ Irreducible jump rates $r_1, r_2 : S \times S \to [0, \infty)$.
  - ⋆ Normalized stationary measures m_1, m_2, i.e, $\max_{x \in S} m_i(x) = 1$.
  - ⋆ $S_\star = \{x \in S : m_1(x) = m_2(x) = 1\}$
- **Configuration space** $\Omega_N = \{\boldsymbol{\eta} = (\eta_1, \eta_2) \in \mathbb{N}_0^S \times \mathbb{N}_0^S : |\eta_1| = A,\ |\eta_2| = B\}$
  - ⋆ $\boldsymbol{\eta} = (\boldsymbol{\eta}(x))_{x \in S},\ \boldsymbol{\eta}(x) = (\eta_1(x), \eta_2(x)) \in \{0, 1, \dots, A\} \times \{0, 1, \dots, B\}$
  - ⋆ $\rho_N := A/N \to \rho \in (0, 1)$
- **Dynamics** Species i particle jumps $x \to y$ at rate $\boldsymbol{g}_i(\boldsymbol{\eta}(x)) r_i(x, y)$

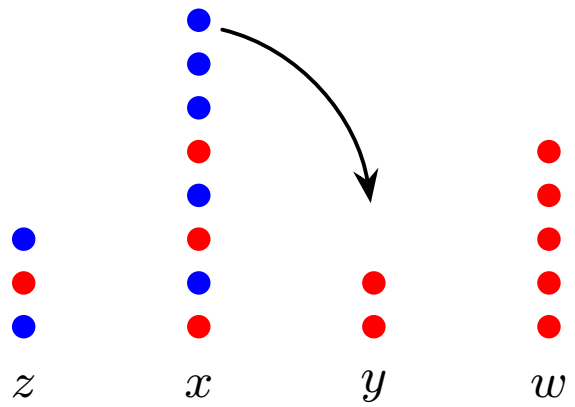
$$\boldsymbol{g}_i(k_1, k_2) = \begin{cases} \frac{k_i}{k_1 + k_2} g(k_1 + k_2) & \text{if } k_1 + k_2 \geq 1 \\ 0 & \text{if } k_1 + k_2 = 0 \end{cases}$$



Two-species zero-range process

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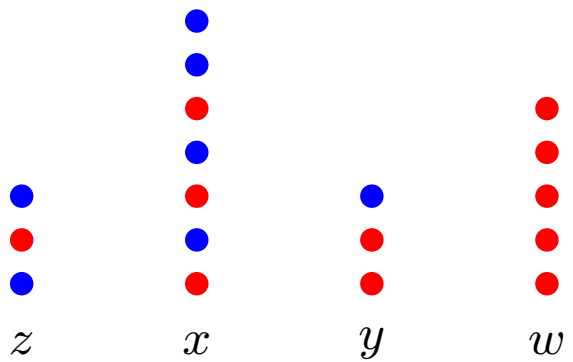
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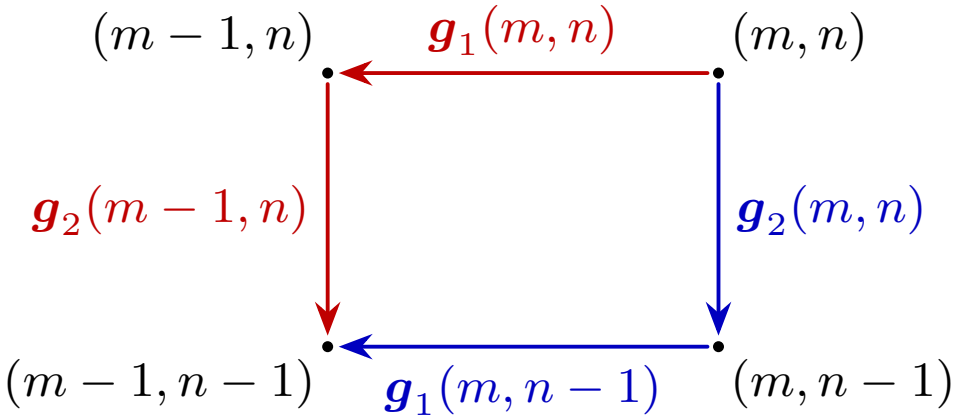


Stationary distribution ν_N

- **Stationary distribution** Product stationary state $\Longleftrightarrow$ compatibility relation:

$$\boldsymbol{g}_1(m,n-1)\boldsymbol{g}_2(m,n) = \boldsymbol{g}_2(m-1,n)\boldsymbol{g}_1(m,n), \quad m,n \geq 1.$$

Evans–Hanney *J. Phys. A* '03



- **Stationary distribution** With convention $\boldsymbol{g!}(0,0) = 1$,

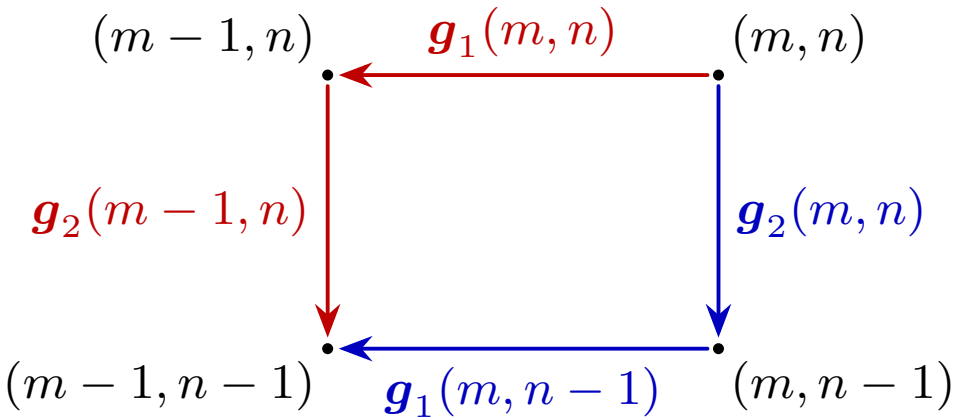
$$\nu_N(\boldsymbol{\eta}) = \frac{N^\alpha}{Z_N \binom{N}{A}} \prod_{x \in S} \frac{m_1(x)^{\eta_1(x)} m_2(x)^{\eta_2(x)}}{\boldsymbol{g!}(\boldsymbol{\eta}(x))}, \quad \boldsymbol{g!}(m,n) = \prod_{p=1}^{m} \boldsymbol{g}_1(p,0) \prod_{q=1}^{n} \boldsymbol{g}_2(m,q).$$

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Condensation

- **Metastable well** For each $x \in S$ and $1 \ll \ell_N \ll N$,

$$\mathcal{E}_N^x := \{\boldsymbol{\eta} \in \Omega_N : |\boldsymbol{\eta}(x)| \geq N - \ell_N\}, \quad |\eta(x)| := \eta_1(x) + \eta_2(x)$$

Theorem (Kim–L. arXiv : 2609.03821)

1. *For $\alpha \in (1, \infty)$, we have*

$$\lim_{N\to\infty} \nu_N(\mathcal{E}_N^x) = \frac{1}{|S_\star|} \quad (x \in S_\star) \quad \text{and} \quad \lim_{N\to\infty} \nu_N(\Delta_N) = 0. \tag{2}$$

2. *If $\alpha = 1$, (2) also holds if ℓ_N additionally satisfies*

$$\lim_{N\to\infty} \frac{\log \ell_N}{\log N} = 1.$$

3. *If $\alpha \in (0, 1)$, there is no condensation phenomena, i.e.,*

$$\lim_{N\to\infty} \nu_N(\mathcal{E}_N^x) = 0, \quad \text{for all } x \in S.$$

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Main theorem

- **Metastable time scale** $\theta_N = N^{\alpha+1}$.
- **Limiting Markov chain** $\mathbb{X}$ **on** $S_\star$

$$\mathfrak{r}(x,y) = \frac{1}{\Gamma(\alpha)\mathcal{I}_\alpha}\left(\frac{\rho}{\mathrm{cap}_1(x,y)} + \frac{1-\rho}{\mathrm{cap}_2(x,y)}\right)^{-1}, \quad x,y \in S_\star.$$

Theorem (Kim–L. arXiv : 2609.03821)

For sufficiently small $\gamma > 0$, *let* $\ell_N = \lfloor \gamma \log N \rfloor$. *Then for* $\alpha > 1$ *and for all* $x \in S_\star$,

1. ***(M) Local metastability*** *For* $b \in (0,1)$

$$\lim_{N\to\infty} \inf_{\boldsymbol{\eta}\in\mathcal{E}_N^x} \mathbb{P}_{\boldsymbol{\eta}}^N \left[H_{\boldsymbol{\xi}_N^x} < H_{\Omega_N \setminus \tilde{\mathcal{E}}_N^x} \right] = 1, \quad \tilde{\mathcal{E}}_N^x = \{\boldsymbol{\eta} \in \Omega_N : |\boldsymbol{\eta}(x)| \geq N - \lfloor N^b \rfloor\}$$

2. ***(C) Convergence*** $\{X_N(\theta_N t)\}_{t\geq 0} \xrightarrow[N\to\infty]{} \{\mathbb{X}(t)\}_{t\geq 0}$.
3. ***(D) Negligibility***

$$\lim_{N\to\infty} \sup_{\boldsymbol{\eta}\in\mathcal{E}_N^x} \mathbb{E}_{\boldsymbol{\eta}}^N \left[\int_0^T \mathbb{1}\{\boldsymbol{\eta}_N(\theta_N t) \in \Delta_N\} dt \right] = 0, \quad \text{for all} \quad T > 0.$$

Resolvent Approach

- **Metastable description**
  - ⋆ **(C) Convergence** $\{X_N(\theta_N t)\}_{t\geq 0} \xrightarrow[N\to\infty]{} \{\mathbb{X}(t)\}_{t\geq 0}$
  - ⋆ **(D) Negligibility**

$$\lim_{N\to\infty} \sup_{\boldsymbol{\eta}\in\mathcal{E}_N^x} \mathbb{E}_{\boldsymbol{\eta}}^N \left[\int_0^T \mathbb{1}\{\boldsymbol{\eta}_N(\theta_N t) \in \Delta_N\} dt \right] = 0$$

- **Resolvent Approach**: $\forall\, \mathbf{g} : S_\star \to \mathbb{R}$, $\lambda > 0$,

$$(\lambda - \theta_N \mathcal{L}_N) F_N = \sum_{x\in S_\star} \mathbf{g}(x) \mathbb{1}_{\mathcal{E}_N^x}, \quad (\lambda - \mathfrak{L}_{\mathbb{X}}) \mathfrak{f} = \mathbf{g},$$

$$\mathfrak{R}_{\mathfrak{L}_{\mathbb{X}}} : F_N \simeq \mathfrak{f}(x) \quad \text{on each } \mathcal{E}_N^x \ (x \in S_\star) \implies \textbf{(C)+(D)}$$

Landim–Marcondes–Seo, *J. Eur. Math. Soc.* '25

- $\mathfrak{R}_{\mathfrak{L}_{\mathbb{X}}} \iff \mathfrak{R}^{(1)} + \mathfrak{R}^{(2)}$:
  - ⋆ $\mathfrak{R}^{(1)}$: F_N is asymptotically constant with in each well $\mathcal{E}_N^x$:

$$\lim_{N\to\infty} \sup_{\boldsymbol{\eta},\boldsymbol{\xi}\in\mathcal{E}_N^x} |F_N(\boldsymbol{\eta}) - F_N(\boldsymbol{\xi})| = 0, \quad \forall\, x \in S_\star.$$

  - ⋆ $\mathfrak{R}^{(2)}$: Let $f_N(x) = \mathbb{E}_{\nu_N}[F_N \mid \mathcal{E}_N^x]$, then $f_N(x) \to \mathfrak{f}(x)$.

$\mathfrak{R}^{(1)}$ condition

Let $\boldsymbol{\xi}_N^x = A\delta_x^1 + B\delta_x^2$. It suffices to show that

$$\lim_{N\to\infty} \sup_{\boldsymbol{\eta}\in\mathcal{E}_N^x} |F_N(\boldsymbol{\eta}) - F_N(\boldsymbol{\xi}_N^x)| = 0, \quad \text{for each } x \in S_\star.$$

Let $H_{\boldsymbol{\xi}_N^x}$ be the $\boldsymbol{\xi}_N^x$-hitting time.

$$F_N(\boldsymbol{\eta}) = \mathbb{E}_{\boldsymbol{\eta}}^N \left[\int_0^\infty e^{-\lambda t} G_N(\boldsymbol{\eta}(\theta_N t)) dt \right], \quad G_N = \sum_{x\in S_\star} \mathbf{g}(x) \mathbb{1}_{\mathcal{E}_N^x}.$$

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Let $H_{\boldsymbol{\xi}_N^x}$ be the $\boldsymbol{\xi}_N^x$-hitting time. With change of variables and strong Markov property:

$$F_N(\boldsymbol{\eta}) = \theta_N^{-1}\mathbb{E}_{\boldsymbol{\eta}}^N \left[\int_0^{H_{\boldsymbol{\xi}_N^x}} e^{-\lambda\theta_N^{-1}t} G_N(\boldsymbol{\eta}_N(t))dt\right] + \mathbb{E}_{\boldsymbol{\eta}}^N \left[e^{-\lambda\theta_N^{-1}H_{\boldsymbol{\xi}_N^x}}\right] F_N(\boldsymbol{\xi}_N^x).$$

Thus,

$$|F_N(\boldsymbol{\eta}) - F_N(\boldsymbol{\xi}_N^x)| \le 2\theta_N^{-1}\|\mathbf{g}\|_\infty \mathbb{E}_{\boldsymbol{\eta}}^N \left[H_{\boldsymbol{\xi}_N^x}\right]$$

Therefore, it suffices to show that

$$\sup_{\boldsymbol{\eta}\in\mathcal{E}_N^x} \mathbb{E}_{\boldsymbol{\eta}}^N \left[H_{\boldsymbol{\xi}_N^x}\right] \ll \theta_N$$

Construct a canonical path:

$$\boldsymbol{\eta} = \boldsymbol{\eta}_0, \boldsymbol{\eta}_1, \ldots, \boldsymbol{\eta}_k = \boldsymbol{\xi}_N^x, \quad k \le \kappa\ell_N = O(\log N).$$

$\mathfrak{R}^{(2)}$: idea

Key idea for $\mathfrak{R}^{(2)}$: Test the resolvent equation against $\mathfrak{H}_x^* := \mathfrak{H}^*_{\mathcal{E}_N^x, \mathcal{E}_N \setminus \mathcal{E}_N^x}$

$$\begin{cases} \mathcal{L}_N^* \mathfrak{H}_x^* \equiv 0 & \text{on} \quad \Omega_N \setminus \mathcal{E}_N \\ \mathfrak{H}_x^* \equiv 1 & \text{on} \quad \mathcal{E}_N^x \\ \mathfrak{H}_x^* \equiv 0 & \text{on} \quad \mathcal{E}_N \setminus \mathcal{E}_N^x \end{cases} .$$

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Key idea for $\mathfrak{R}^{(2)}$: Test the resolvent equation against $\mathfrak{H}_x^* := \mathfrak{H}^*_{\mathcal{E}_N^x, \mathcal{E}_N \setminus \mathcal{E}_N^x}$

$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \langle G_N, \mathfrak{H}_x^* \rangle_{\nu_N}$$

Definition of $\mathfrak{H}_x^*$ + condensation:

$$\lambda \langle F_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \lambda f_N(x) \nu_N(\mathcal{E}_N^x) + o(1), \quad \text{and} \quad \langle G_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x).$$

Only remains $\theta_N \langle -\mathcal{L}_N F_N, \mathfrak{H}_x^* \rangle_{\nu_N}$

$$\begin{aligned}
\theta_N \langle F_N, -\mathcal{L}_N^* \mathfrak{H}_x^* \rangle_{\nu_N} &= \theta_N \sum_{\boldsymbol{\eta} \in \mathcal{E}_N} F_N(\boldsymbol{\eta}) (-\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta}) \\
&\overset{\mathfrak{R}^{(1)}}{\simeq} \theta_N f_N(x) \sum_{\boldsymbol{\eta} \in \mathcal{E}_N^x} (-\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta}) \\
&\quad + \theta_N \sum_{y \in S_\star \setminus \{x\}} f_N(y) \sum_{\boldsymbol{\eta} \in \mathcal{E}_N^y} (-\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta})
\end{aligned}$$

$\mathfrak{R}^{(2)}$: idea

Key idea for $\mathfrak{R}^{(2)}$: Test the resolvent equation against $\mathfrak{H}_x^* := \mathfrak{H}^*_{\mathcal{E}_N^x, \mathcal{E}_N \setminus \mathcal{E}_N^x}$

$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \langle G_N, \mathfrak{H}_x^* \rangle_{\nu_N}$$

Definition of $\mathfrak{H}_x^*$ + condensation:

$$\lambda \langle F_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \lambda f_N(x) \nu_N(\mathcal{E}_N^x) + o(1), \quad \text{and} \quad \langle G_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x).$$

Only remains $\theta_N \langle -\mathcal{L}_N F_N, \mathfrak{H}_x^* \rangle_{\nu_N}$

$$\begin{aligned}
\theta_N \langle F_N, -\mathcal{L}_N^* \mathfrak{H}_x^* \rangle_{\nu_N} &= \theta_N \sum_{\boldsymbol{\eta} \in \mathcal{E}_N} F_N(\boldsymbol{\eta}) (-\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta}) \\
&\overset{\mathfrak{R}^{(1)}}{\simeq} \theta_N f_N(x) \sum_{y \in S_\star \setminus \{x\}} \sum_{\boldsymbol{\eta} \in \mathcal{E}_N^y} (\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta}) \\
&\quad + \theta_N \sum_{y \in S_\star \setminus \{x\}} f_N(y) \sum_{\boldsymbol{\eta} \in \mathcal{E}_N^y} (-\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta})
\end{aligned}$$

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$$\begin{aligned} \theta_N \langle F_N, -\mathcal{L}_N^* \mathfrak{H}_x^* \rangle_{\nu_N} &= \theta_N \sum_{\boldsymbol{\eta} \in \mathcal{E}_N} F_N(\boldsymbol{\eta}) (-\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta}) \\ &\overset{\mathfrak{R}^{(1)}}{\simeq} \sum_{y \in S_\star \setminus \{x\}} \theta_N \Bigg(\sum_{\boldsymbol{\eta} \in \mathcal{E}_N^y} (\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta}) \Bigg) [f_N(x) - f_N(y)] \end{aligned}$$

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&\overset{\mathfrak{R}^{(1)}}{\simeq} \sum_{y \in S_\star \setminus \{x\}} \nu_N(\mathcal{E}_N^x) \mathfrak{r}(x,y) [f_N(x) - f_N(y)] \\
&\simeq -\nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x)
\end{aligned}$$

Then,

$$(\lambda - \mathfrak{L}_{\mathbb{X}}) f_N(x) \simeq \mathbf{g}(x) \ \ (x \in S_\star) \implies (\lambda - \mathfrak{L}_{\mathbb{X}})(f_N - \mathfrak{f}) \simeq 0.$$

However, $\mathfrak{H}_x^*$ is **unknown.**

$\mathfrak{R}^{(2)}$: idea

Key idea for $\mathfrak{R}^{(2)}$: Test the resolvent equation against $\mathfrak{H}_x^* := \mathfrak{H}^*_{\mathcal{E}_N^x, \mathcal{E}_N \setminus \mathcal{E}_N^x}$

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&\simeq -\nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x)
\end{aligned}$$

Approximate $\mathfrak{H}_x^*$ by $\mathbb{W}_x$:

$$\mathbb{W}_x|_{\mathcal{E}_N^x} \equiv 1, \quad \mathbb{W}_x|_{\mathcal{E}_N \setminus \mathcal{E}_N^x} \equiv 0, \quad \theta_N \langle -\mathcal{L}_N F_N, \mathfrak{H}_x^* \rangle_{\nu_N} \simeq \theta_N \langle -\mathcal{L}_N F_N, \mathbb{W}_x \rangle_{\nu_N}$$

H^1-approximation

- First,

$$\begin{aligned}\sup_{N\geq 1}\theta_N D_N(F_N) &= \sup_{N\geq 1}\sum_{\boldsymbol{\eta}\in\Omega_N} F_N(\boldsymbol{\eta})(-\theta_N\mathcal{L}_N F_N)(\boldsymbol{\eta})\nu_N(\boldsymbol{\eta})\\ &= \sup_{N\geq 1}\sum_{\boldsymbol{\eta}\in\Omega_N} F_N(\boldsymbol{\eta})(G_N-\lambda F_N)\nu_N(\boldsymbol{\eta})\\ &\leq \sup_{N\geq 1}\|F_N\|_\infty\|G_N-\lambda F_N\|_\infty<\infty.\end{aligned}$$

- **Sector condition** $\langle G,-\mathcal{L}_N F\rangle_{\nu_N}^2 \lesssim D_N(G)D_N(F)$.

- Together,

$$|\theta_N\langle F_N,-\mathcal{L}_N^*(\mathfrak{H}_x^*-\mathbb{W}_x)\rangle_{\nu_N}|^2 \lesssim \theta_N^2 D_N(F_N)D_N(\mathfrak{H}_x^*-\mathbb{W}_x).$$

- Approximate $\mathbb{W}_x\approx\mathfrak{H}_x^*$

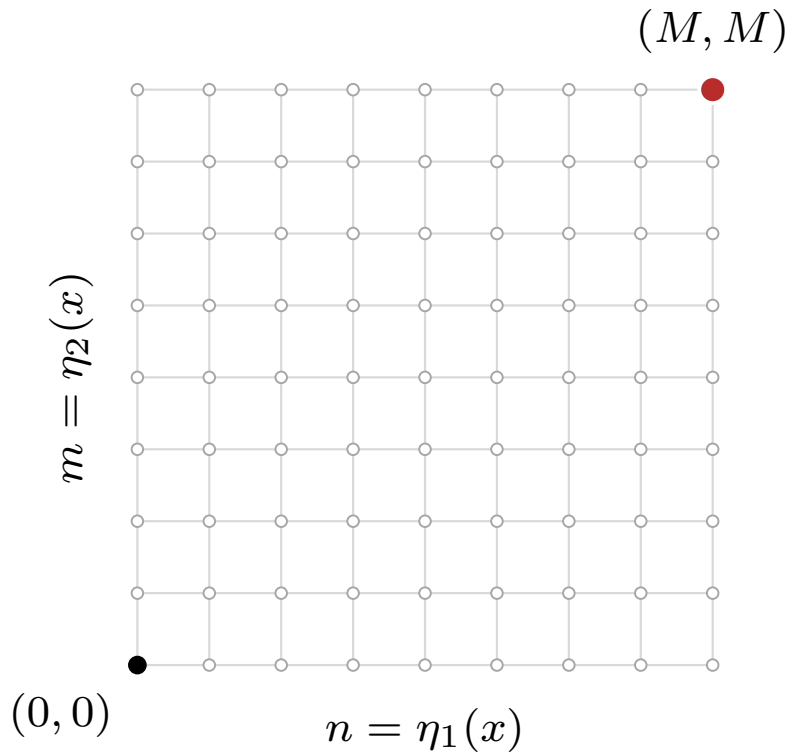
$$\theta_N D_N(\mathfrak{H}_x^*-\mathbb{W}_x)=o(1),$$

(in H^1 sense, Kim–Seo *Ann. Probab.* '25), we can replace $\mathfrak{H}_x^*$ into $\mathbb{W}_x$!

$\mathfrak{R}^{(2)}$: Reduced model

Two sites $S = \{x, y\}$, M particles of each species and $N = 2M$.

$$(n, m) = (\eta_1(x), \eta_2(x)), \quad \eta(y) = (M - n, M - m), \quad \mathcal{E}_N^z = \{\boldsymbol{\xi}_N^z\}, \ (z \in \{x, y\}).$$



Two condensates

Both species are at x:

$$(n, m) = (M, M).$$

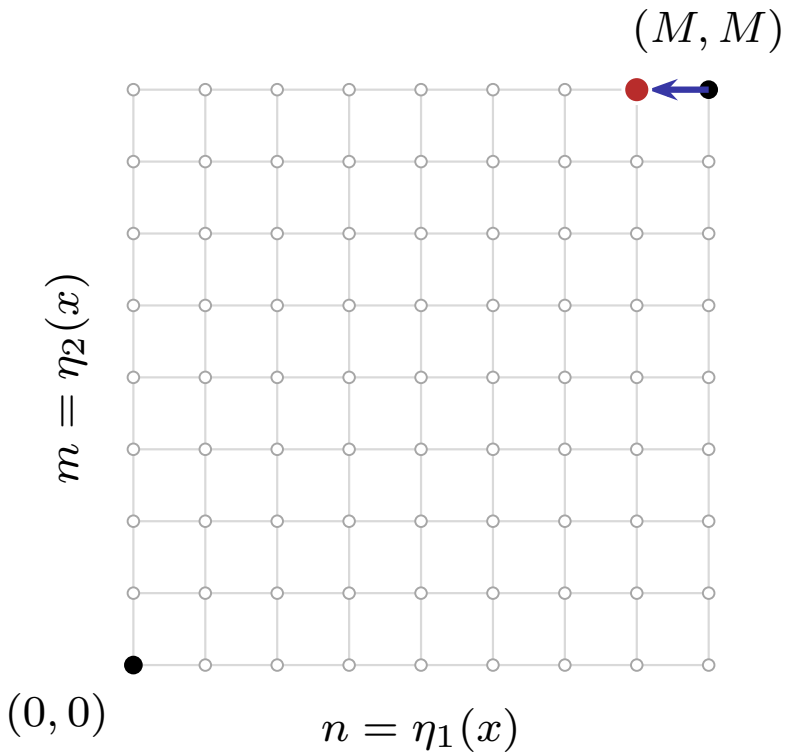
The other corner $(0, 0)$ represents condensation at y.

We seek a test function equal to 1 and 0 at these corners.

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A single-particle jump

One species-1 particle moves from x to y:

$$(M, M) \longrightarrow (M - 1, M).$$

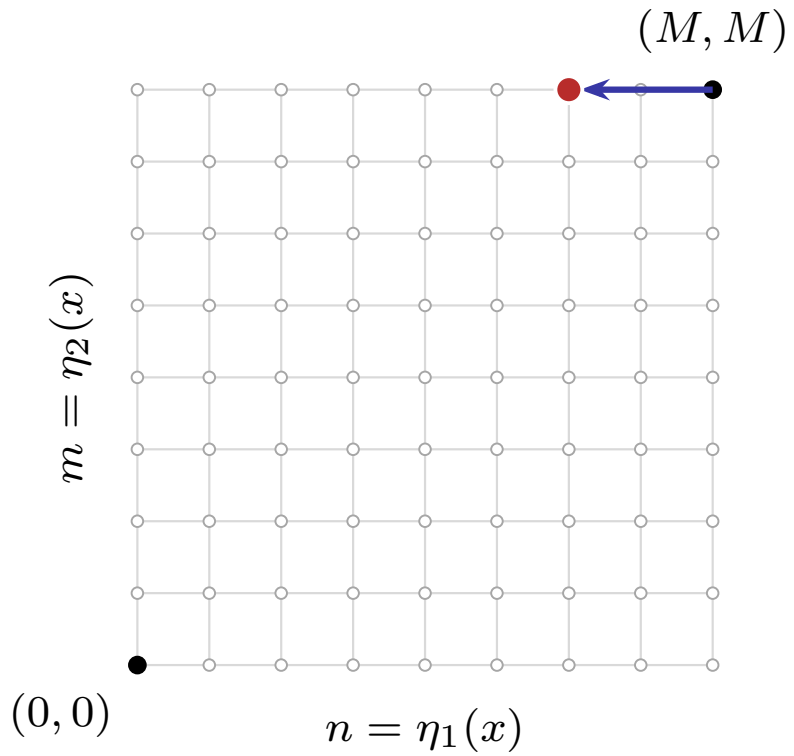
Its jump rate is

$$\tfrac{1}{2} g(2M)\, r_1(x, y).$$

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Two sites $S = \{x, y\}$, M particles of each species and $N = 2M$.

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An off-diagonal excursion

Suppose another species-1 particle leaves x:

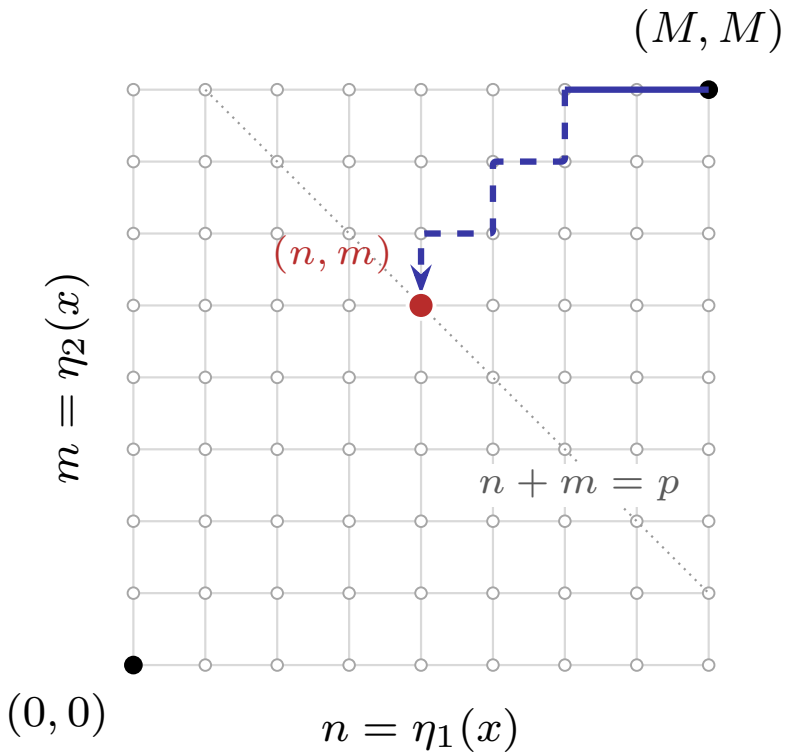
$$(M - 1, M) \longrightarrow (M - 2, M).$$

The species composition at x becomes unbalanced.

$\mathfrak{R}^{(2)}$: Reduced model

Two sites $S = \{x, y\}$, M particles of each species and $N = 2M$.

$$(n, m) = (\eta_1(x), \eta_2(x)), \quad \eta(y) = (M - n, M - m), \quad \mathcal{E}_N^z = \{\boldsymbol{\xi}_N^z\}, \ (z \in \{x, y\}).$$



An intermediate configuration

Stop at a state with

$$p = n + m$$

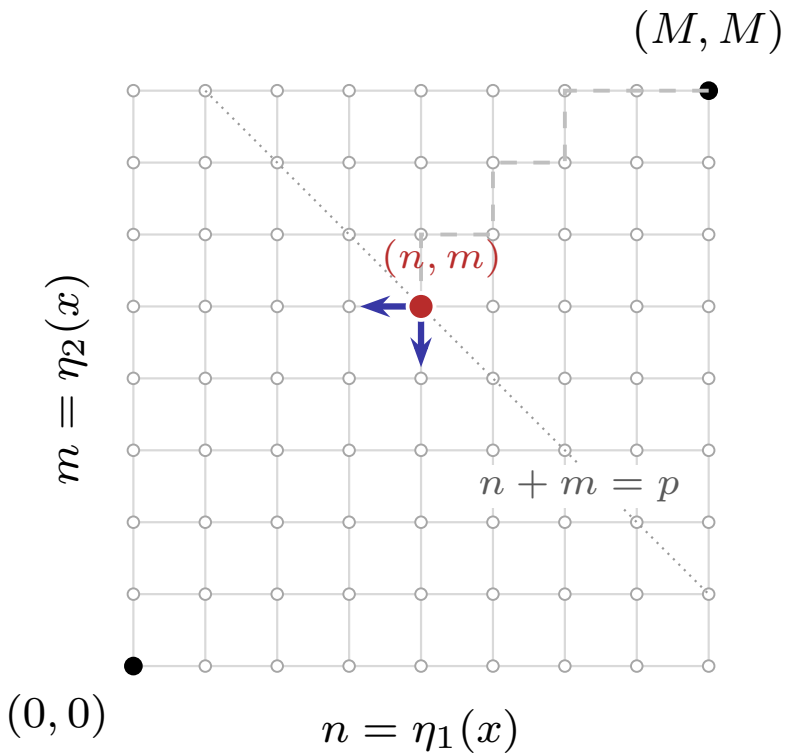
$$\eta(x) = (n, m), \quad \eta(y) = (M-n, M-m).$$

The dashed segment is an illustrative sequence of $x \to y$ jumps.

$\mathfrak{R}^{(2)}$: Reduced model

Two sites $S = \{x, y\}$, M particles of each species and $N = 2M$.

$$(n, m) = (\eta_1(x), \eta_2(x)), \quad \eta(y) = (M - n, M - m), \quad \mathcal{E}_N^z = \{\boldsymbol{\xi}_N^z\}, \ (z \in \{x, y\}).$$



Two ways to transfer one particle

For the $x \to y$ jumps at $p = n + m$,

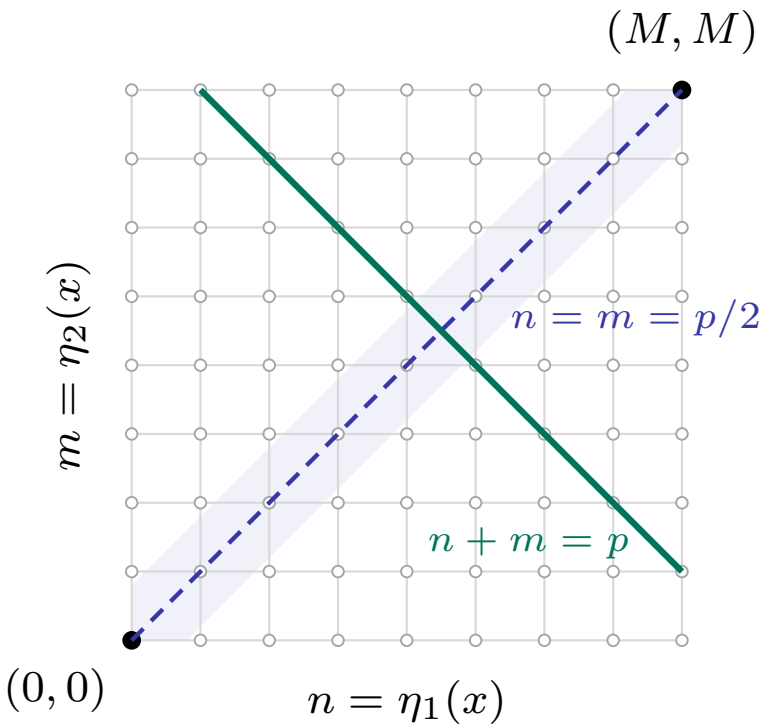
$$q_{\leftarrow} = \frac{n}{p} g(p)\, r_1(x, y),$$

$$q_{\downarrow} = \frac{m}{p} g(p)\, r_2(x, y).$$

Both lower p by one, but change $n - m$ by -1 and $+1$, respectively.

Reduced model: why a composition corrector?

$$p = n + m, \qquad \rho = \tfrac{1}{2}, \qquad R_i = r_i(x, y) + r_i(y, x).$$



A total-mass profile

The one-dimensional energy estimate suggests

$$H_p = \frac{\sum_{k=0}^{p-1} a(k)a(N-k-1)}{\sum_{k=0}^{N-1} a(k)a(N-k-1)}.$$

Thus $H_0 = 0$ and $H_N = 1$.

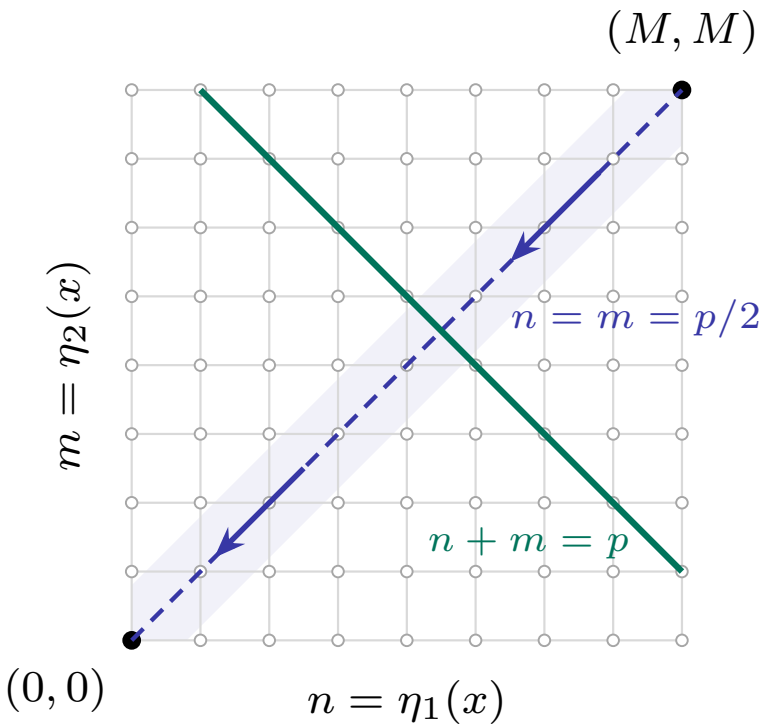
If $R_1 \neq R_2$, optimal increments in the two species directions differ:

$$\nabla_1 G : \nabla_2 G \approx R_1^{-1} : R_2^{-1}.$$

A function of p alone has equal increments in both directions.

Reduced model: why a composition corrector?

$$p = n + m, \qquad \rho = \tfrac{1}{2}, \qquad R_i = r_i(x, y) + r_i(y, x).$$



Conditional composition

Under the stationary law, given $n + m = p$,

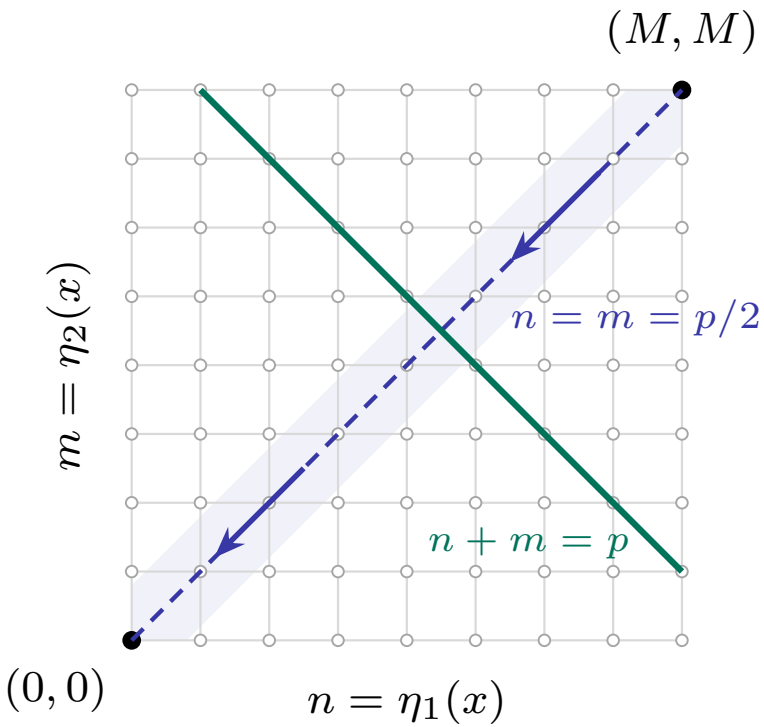
$$\nu_N(n, p - n \mid p) = \frac{\binom{p}{n}\binom{N-p}{M-n}}{\binom{N}{M}}.$$

Deviation

$$n - \frac{p}{2} = \frac{n - m}{2}, \quad \mathbb{E}[n \mid p] = \frac{p}{2}.$$

Reduced model: why a composition corrector?

$$p = n + m, \qquad \rho = \tfrac{1}{2}, \qquad R_i = r_i(x,y) + r_i(y,x).$$



Correcting for unequal rates

$$\boxed{G(n,m) = H_p + \beta_p \left(n - \frac{p}{2} \right)}$$

$$\beta_p = \frac{H_p - H_{p-1}}{\vartheta} \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad \beta_0 = 0,$$

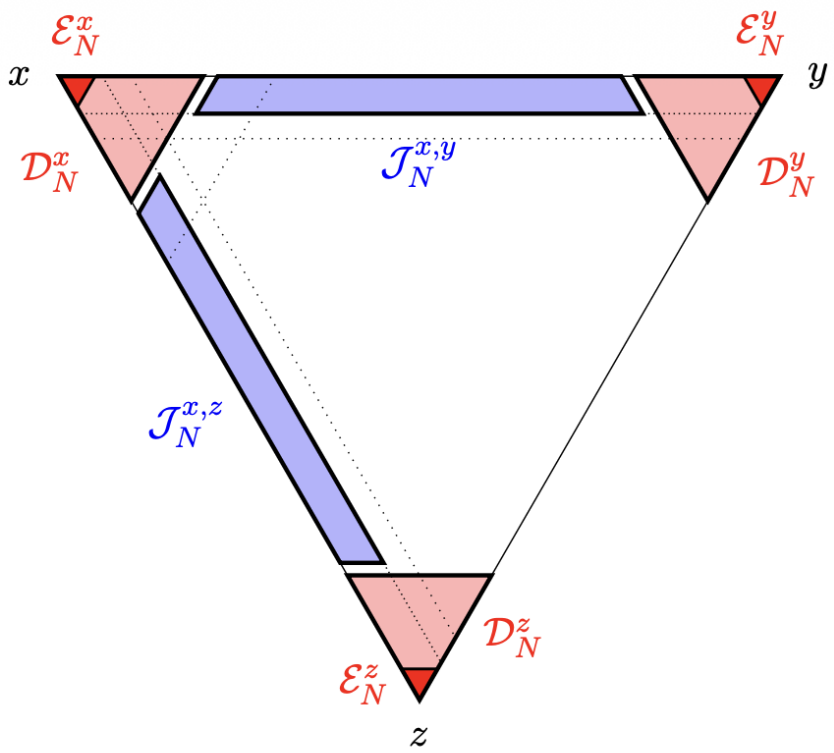
$$\vartheta = \frac{1}{2R_1} + \frac{1}{2R_2}.$$

The correction has zero conditional mean and adjusts the two directional increments.

This is the prototype the test function on the tube.

Wells, tubes, and the global test function

Choose $\pi_N = \lfloor N^\beta \rfloor$ where $\beta \in (\frac{1}{\alpha}, 1)$.



- **Enlarged wells**

$$\mathcal{D}_N^z := \{\boldsymbol{\eta} : |\boldsymbol{\eta}(z)| \geq N - \pi_N\}, \ z \in S_\star$$

- **Tubes** For $x \neq y \in S_\star$

$$\mathcal{T}_N^{x,y} = \{\boldsymbol{\eta} : |\boldsymbol{\eta}(x)| + |\boldsymbol{\eta}(y)| \geq N - \pi_N\},$$

$$\mathcal{J}_N^{x,y} = \mathcal{T}_N^{x,y} \setminus (\mathcal{D}_N^x \cup \mathcal{D}_N^y).$$

- **Good region**

$$\mathcal{G}_N^x := \bigcup_{z \in S_\star} \mathcal{D}_N^z \cup \bigcup_{y \in S_\star \setminus \{x\}} \mathcal{J}_N^{x,y}$$

$$\mathbb{W}_x(\boldsymbol{\eta}) = \sum_{y \in S_\star \setminus \{x\}} \Theta_\epsilon^{x,y}(\boldsymbol{\eta}/N) \mathcal{W}_{x,y}(\boldsymbol{\eta}),$$

Test function on a saddle tube

Macroscopic coordinate

For $x \neq y \in S_\star$,

$$\mathcal{X}^i_{x,y}(\boldsymbol{\eta}) := \frac{1}{N} \sum_{u \in \{x,y\}} \eta_i(u) \mathfrak{h}^*_{i,x,y}(u) + \min \left\{ \frac{1}{N} \sum_{u \in S \setminus \{x,y\}} \eta_i(u) \mathfrak{h}^*_{i,x,y}(u), \epsilon \right\} \approx \frac{\eta_i(x)}{N}$$

Test function on tube

For $u = \mathcal{X}^1_{x,y}$, $w = \mathcal{X}^2_{x,y}$,

$$\mathcal{W}_{x,y}(\boldsymbol{\eta}) = \underbrace{H_\epsilon(u+w)}_{\text{one-dimensional profile}} + \underbrace{\omega_N(x,y) H'_\epsilon(u+w)(u - \rho_N(u+w))}_{\text{corrector}}.$$

And the coefficients:

$$\tau_N(x,y) = \frac{\rho_N}{\mathrm{cap}_1(x,y)} + \frac{1-\rho_N}{\mathrm{cap}_2(x,y)}, \quad \omega_N(x,y) = \frac{\mathrm{cap}_1(x,y)^{-1} - \mathrm{cap}_2(x,y)^{-1}}{\tau_N(x,y)}.$$

If $\mathrm{cap}_1(x,y) = \mathrm{cap}_2(x,y)$, the corrector vanishes.

Test function on a saddle tube

Macroscopic coordinate

For $x \neq y \in S_\star$,

$$\mathcal{X}^i_{x,y}(\boldsymbol{\eta}) := \frac{1}{N} \sum_{u \in \{x,y\}} \eta_i(u) \mathfrak{h}^*_{i,x,y}(u) + \min \left\{ \frac{1}{N} \sum_{u \in S \setminus \{x,y\}} \eta_i(u) \mathfrak{h}^*_{i,x,y}(u), \epsilon \right\} \approx \frac{\eta_i(x)}{N}$$

Test function on tube

For $u = \mathcal{X}^1_{x,y}$, $w = \mathcal{X}^2_{x,y}$,

$$\mathcal{W}_{x,y}(\boldsymbol{\eta}) = \underbrace{H_\epsilon(u+w)}_{\text{one-dimensional profile}} + \underbrace{\omega_N(x,y) H'_\epsilon(u+w)(u - \rho_N(u+w))}_{\text{corrector}}.$$

$$\mathbb{W}_x|_{\mathcal{D}^x_N} \equiv 1, \quad \mathbb{W}_x|_{\mathcal{D}^z_N} \equiv 0, (z \in S_\star \setminus \{x\}), \quad \mathbb{W}_x|_{\mathcal{J}^{x,y}_N} \equiv \mathcal{W}_{x,y}$$

$\mathfrak{R}^{(2)}$: Testing with $\mathbb{W}_x$

$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathbb{W}_x \rangle_{\nu_N} = \langle G_N, \mathbb{W}_x \rangle_{\nu_N}.$$

- $\mathbb{W}_x$ definition + condensation

$$\lambda f_N(x) \nu_N(\mathcal{E}_N^x) + \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x) + o(1)$$

- Only remains $\theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N}$,

$$\theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} = \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{G}_N^x} + \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, (\mathcal{G}_N^x)^c}.$$

- **Motivation** For $\mathcal{E} \subset \Omega_N$, $\langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{E}}$

$$\sum_{i \in \{1,2\}} \sum_{\boldsymbol{\eta} \in \mathcal{E}} \sum_{\substack{w \in S \\ u : \eta_i(u) \geq 1}} \nu_N(\boldsymbol{\eta}) \boldsymbol{g}_i(\boldsymbol{\eta}(u)) r_i^*(u, w) F_N(\boldsymbol{\eta}) [\mathbb{W}_x(\boldsymbol{\eta} - \delta_u^i + \delta_w^i) - \mathbb{W}_x(\boldsymbol{\eta})]$$

change $\boldsymbol{\eta} - \delta_u^i \to \boldsymbol{\xi} \in \Omega_{N-1}^{i,-}$

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$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathbb{W}_x \rangle_{\nu_N} = \langle G_N, \mathbb{W}_x \rangle_{\nu_N}.$$

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$$\lambda f_N(x) \nu_N(\mathcal{E}_N^x) + \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x) + o(1)$$

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- **Motivation** For $\mathcal{E} \subset \Omega_N$, $\langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{E}}$

$$\sum_{\boldsymbol{\xi} \in \Omega_{N-1}^{i,-}} \sum_{\substack{u,w \in S \\ \boldsymbol{\xi} + \delta_u^i \in \mathcal{E}}} \frac{\boldsymbol{m}^{\boldsymbol{\xi}}}{\boldsymbol{g!}(\boldsymbol{\xi})} m_i(u) r_i^*(u,w) F_N(\boldsymbol{\xi} + \delta_u^i) [\mathbb{W}_x(\boldsymbol{\xi} + \delta_w^i) - \mathbb{W}_x(\boldsymbol{\xi} + \delta_u^i)]$$

where

$$\Omega_{N-1}^{i,-} := \{ \boldsymbol{\xi} \in \mathbb{N}_0^S \times \mathbb{N}_0^S : |\xi_1| = A - \mathbb{1}_1(i),\ |\xi_2| = B - \mathbb{1}_2(i) \}$$

$\mathfrak{R}^{(2)}$: Testing with $\mathbb{W}_x$

$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathbb{W}_x \rangle_{\nu_N} = \langle G_N, \mathbb{W}_x \rangle_{\nu_N}.$$

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- **Motivation** For $\mathcal{E} \subset \Omega_N$, $\langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{E}}$

$$\sum_{i \in \{1,2\}} \sum_{\boldsymbol{\xi}} \frac{\boldsymbol{m}^{\boldsymbol{\xi}}}{\boldsymbol{g}!(\boldsymbol{\xi})} \sum_{u,w \in S} m_i(u) r_i^*(u,w) [\mathbb{W}_x(\boldsymbol{\xi} + \delta_w^i) - \mathbb{W}_x(\boldsymbol{\xi} + \delta_u^i)] = 0$$

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$$\lambda f_N(x) \nu_N(\mathcal{E}_N^x) + \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x) + o(1)$$

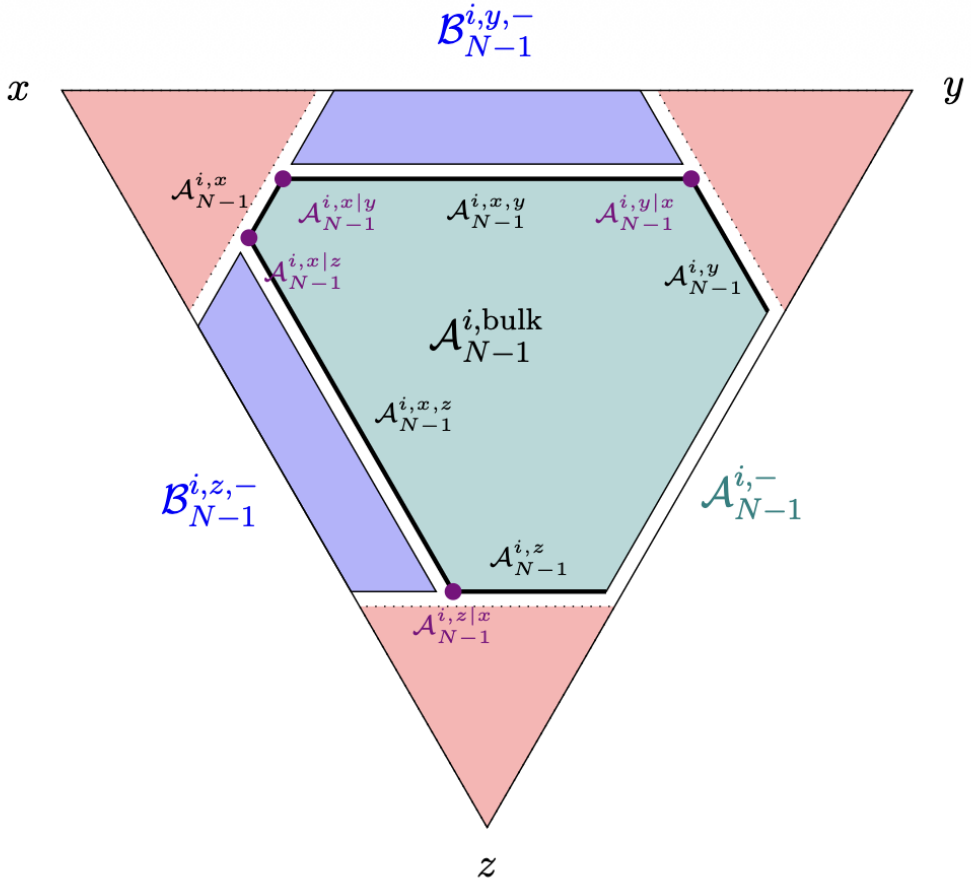
- Only remains $\theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N}$,

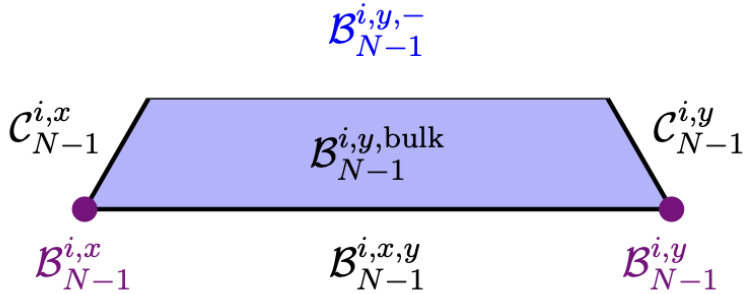
$$\theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} = \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{G}_N^x} + \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, (\mathcal{G}_N^x)^c}.$$

- **Motivation** For $\mathcal{E} \subset \Omega_N$, $\langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{E}}$

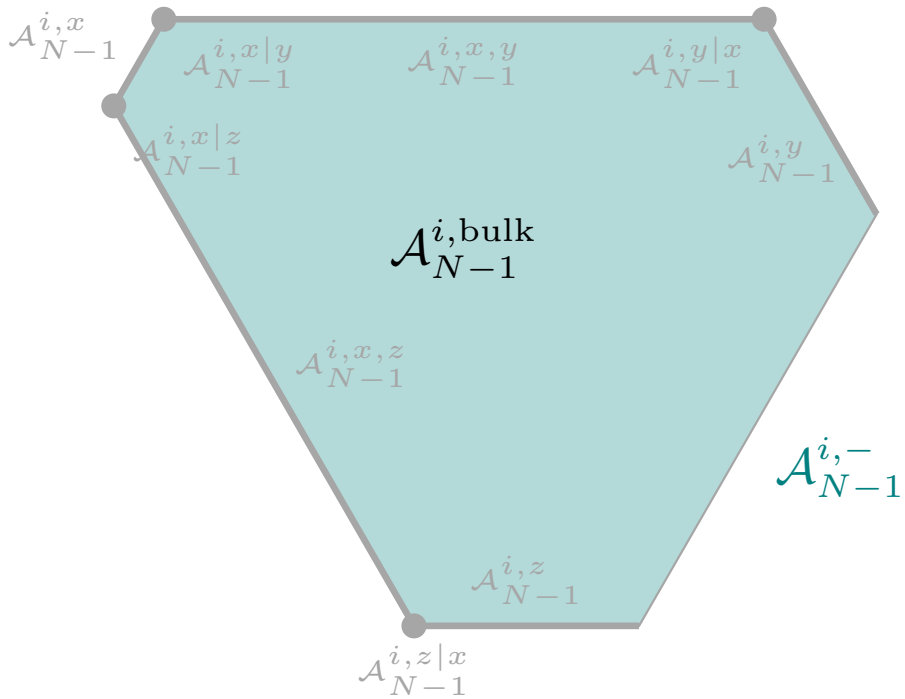
$$\sum_{i \in \{1,2\}} \sum_{\boldsymbol{\xi}} \frac{\boldsymbol{m}^{\boldsymbol{\xi}}}{\boldsymbol{g}!(\boldsymbol{\xi})} \sum_{u,w \in S} m_i(u) r_i^*(u,w) \left(F_N(\boldsymbol{\xi} + \delta_u^i) - \frac{1}{|S|} \sum_{u \in S} F_N(\boldsymbol{\xi} + \delta_u^i) \right) \times [\mathbb{W}_x(\boldsymbol{\xi} + \delta_w^i) - \mathbb{W}_x(\boldsymbol{\xi} + \delta_u^i)]$$

$N-1$ decomposition





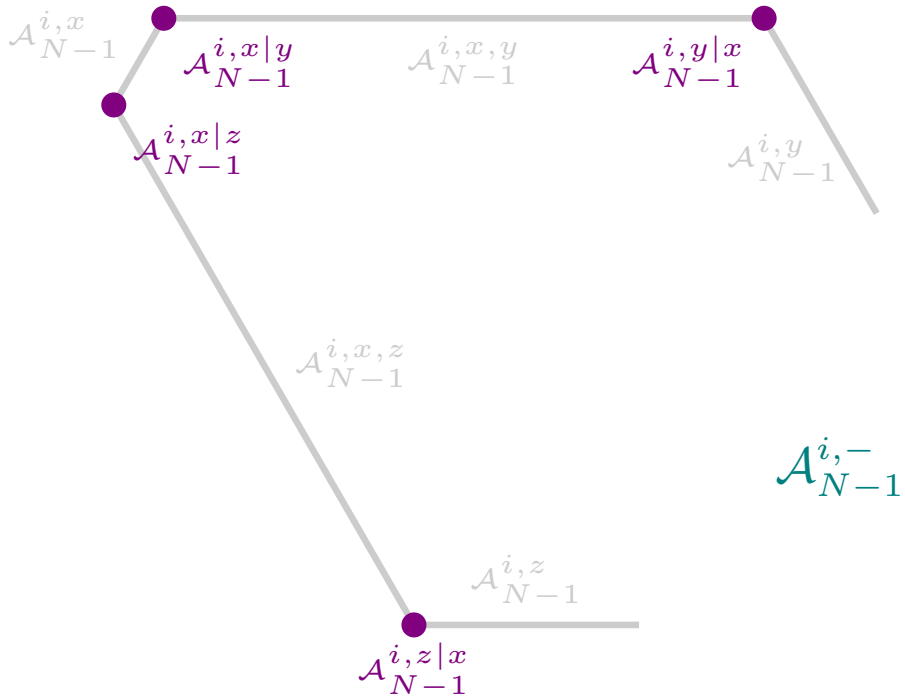
$N-1$ decomposition



Outside bulk Centering and Cauchy–Schwarz give

$$|\text{outside bulk term}| \lesssim \underbrace{\sqrt{\theta_N D_N(F_N)}}_{O(1)} \underbrace{\sqrt{\theta_N D_N(\mathbb{W}_x; (\mathcal{G}_N^x)^c)}}_{o_N(1)}.$$

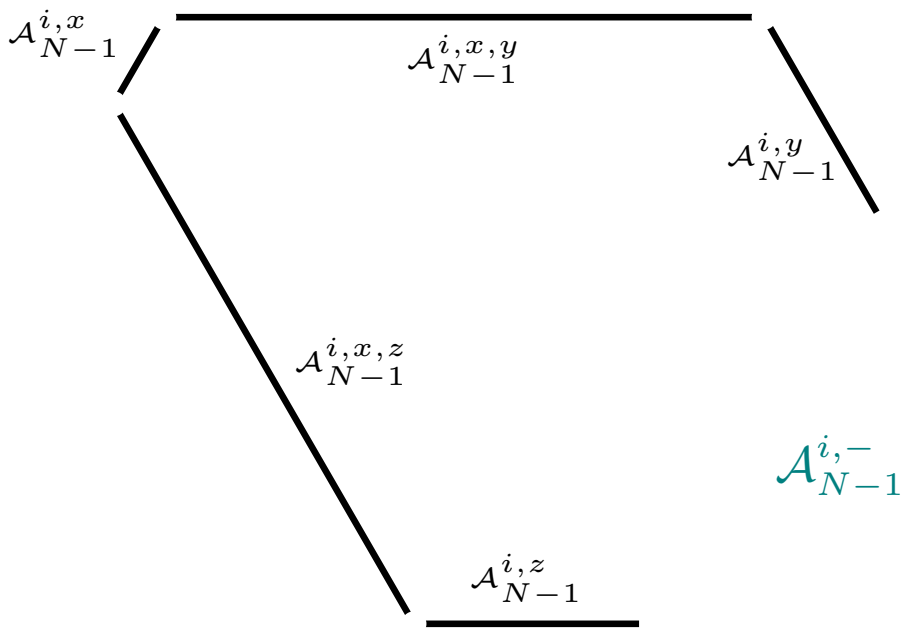
$N-1$ decomposition



Edge adjacent to wells Since $\mathbb{W}_x$ is constant

$$\text{contribution} = 0$$

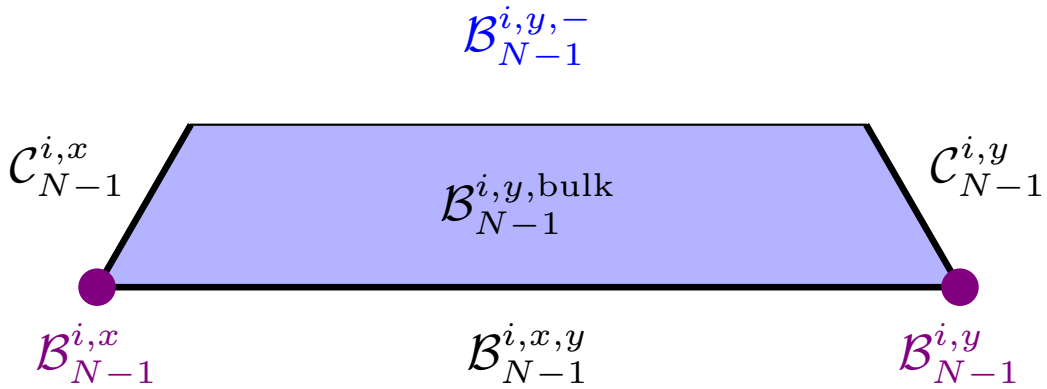
$N-1$ decomposition



Remaining edge For fixed ϵ,

$$|\text{remaining edge}| \;\le\; C_\epsilon \frac{N}{\pi_N^\alpha} \;=\; o_N(1), \qquad N \ll \pi_N^\alpha.$$

$N-1$ decomposition



Region	Contribution	Main reason
Edge adjacent to wells	0	$\mathbb{W}_x$ is constant
Remaining edge	$o_N(1)$	Small stationary mass
Tube bulk $\mathcal{B}_{N-1}^{i,y,\mathrm{bulk}}$	Leading term	Next slide

Leading contribution on a tube

Fix $y \in S_\star \setminus \{x\}$ and $\boldsymbol{\xi} \in \mathcal{B}_{N-1}^{i,y,\mathrm{bulk}} \implies \boldsymbol{\xi} + \delta_u^i, \boldsymbol{\xi} + \delta_w^i \in \mathcal{J}_N^{x,y}$

$$- \sum_{u,w \in S} m_i(u) r_i^*(u,w) F_N(\boldsymbol{\xi} + \delta_u^i)[\mathbb{W}_x(\boldsymbol{\xi} + \delta_w^i) - \mathbb{W}_x(\boldsymbol{\xi} + \delta_u^i)]$$

Leading contribution on a tube

Fix $y \in S_\star \setminus \{x\}$ and $\boldsymbol{\xi} \in \mathcal{B}_{N-1}^{i,y,\mathrm{bulk}} \implies \boldsymbol{\xi} + \delta_u^i, \boldsymbol{\xi} + \delta_w^i \in \mathcal{J}_N^{x,y}$

$$- \sum_{u,w \in S} m_i(u) r_i^*(u,w) F_N(\boldsymbol{\xi} + \delta_u^i) [\mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_w^i) - \mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_u^i)]$$

Leading contribution on a tube

Fix $y \in S_\star \setminus \{x\}$ and $\boldsymbol{\xi} \in \mathcal{B}_{N-1}^{i,y,\mathrm{bulk}} \implies \boldsymbol{\xi} + \delta_u^i, \boldsymbol{\xi} + \delta_w^i \in \mathcal{J}_N^{x,y}$

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- **Increment of the test function on the tube**

$$\mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_w^i) - \mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_u^i)$$
$$= \frac{|\boldsymbol{\xi}(x)|^\alpha |\boldsymbol{\xi}(y)|^\alpha}{N^{2\alpha+1} \mathcal{I}_\alpha \tau_N(x,y) \operatorname{cap}_i(x,y)} \left[\mathfrak{h}_{i,x,y}^*(w) - \mathfrak{h}_{i,x,y}^*(u)\right] + \text{negligible term}$$

Leading contribution on a tube

Fix $y \in S_\star \setminus \{x\}$ and $\boldsymbol{\xi} \in \mathcal{B}_{N-1}^{i,y,\mathrm{bulk}} \implies \boldsymbol{\xi} + \delta_u^i, \boldsymbol{\xi} + \delta_w^i \in \mathcal{J}_N^{x,y}$

$$-\frac{|\boldsymbol{\xi}(x)|^\alpha |\boldsymbol{\xi}(y)|^\alpha}{N^{2\alpha+1}\mathcal{I}_\alpha \tau_N(x,y)\,\mathrm{cap}_i(x,y)} \sum_{u,w\in S} m_i(u) r_i^*(u,w)[\mathfrak{h}_{i,x,y}^*(w) - \mathfrak{h}_{i,x,y}^*(u)] F_N(\boldsymbol{\xi} + \delta_u^i)$$

- **Increment of the test function on the tube**

$$\mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_w^i) - \mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_u^i)$$
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$$-\frac{|\boldsymbol{\xi}(x)|^\alpha |\boldsymbol{\xi}(y)|^\alpha}{N^{2\alpha+1}\mathcal{I}_\alpha \tau_N(x,y)\,\mathrm{cap}_i(x,y)} \sum_{u,w\in S} m_i(u) r_i^*(u,w)[\mathfrak{h}_{i,x,y}^*(w) - \mathfrak{h}_{i,x,y}^*(u)] F_N(\boldsymbol{\xi} + \delta_u^i)$$

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$$-\frac{|\boldsymbol{\xi}(x)|^\alpha |\boldsymbol{\xi}(y)|^\alpha}{N^{2\alpha+1}\mathcal{I}_\alpha \tau_N(x,y)\,\mathrm{cap}_i(x,y)} \sum_{u,w\in S} m_i(u) r_i^*(u,w)[\mathfrak{h}_{i,x,y}^*(w) - \mathfrak{h}_{i,x,y}^*(u)] F_N(\boldsymbol{\xi} + \delta_u^i)$$

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$$\mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_w^i) - \mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_u^i)$$
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- **Harmonicity** Since $m_i(x) = m_i(y) = 1$,

$$L_i^* \mathfrak{h}_{i,x,y}^* = \mathrm{cap}_i(x,y)\left(\mathbf{1}_{\{y\}} - \mathbf{1}_{\{x\}}\right).$$

Leading contribution on a tube

Fix $y \in S_\star \setminus \{x\}$ and $\boldsymbol{\xi} \in \mathcal{B}_{N-1}^{i,y,\mathrm{bulk}} \implies \boldsymbol{\xi} + \delta_u^i, \boldsymbol{\xi} + \delta_w^i \in \mathcal{J}_N^{x,y}$

$$\frac{|\boldsymbol{\xi}(x)|^\alpha |\boldsymbol{\xi}(y)|^\alpha}{N^{2\alpha+1}\mathcal{I}_\alpha \tau_N(x,y)} [F_N(\boldsymbol{\xi} + \delta_x^i) - F_N(\boldsymbol{\xi} + \delta_y^i)]$$

- **Increment of the test function on the tube**

$$\mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_w^i) - \mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_u^i)$$
$$= \frac{|\boldsymbol{\xi}(x)|^\alpha |\boldsymbol{\xi}(y)|^\alpha}{N^{2\alpha+1}\mathcal{I}_\alpha \tau_N(x,y)\,\mathrm{cap}_i(x,y)} \left[\mathfrak{h}_{i,x,y}^*(w) - \mathfrak{h}_{i,x,y}^*(u)\right] + \text{negligible term}$$

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Fix $y \in S_\star \setminus \{x\}$ and $\boldsymbol{\xi} \in \mathcal{B}_{N-1}^{i,y,\mathrm{bulk}} \implies \boldsymbol{\xi} + \delta_u^i, \boldsymbol{\xi} + \delta_w^i \in \mathcal{J}_N^{x,y}$

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- **Increment of the test function on the tube**

$$\mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_w^i) - \mathcal{W}_{x,y}(\boldsymbol{\xi} + \delta_u^i)$$
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$$L_i^* \mathfrak{h}_{i,x,y}^* = \mathrm{cap}_i(x,y)\big(\mathbf{1}_{\{y\}} - \mathbf{1}_{\{x\}}\big).$$

Conditional averaging

- Fix the configuration $\boldsymbol{\sigma}$ outside $\{x, y\}$, and average over the species composition at fixed total mass on x:

$$\overline{F}_N^{\boldsymbol{\sigma},p} := \mathbb{E}_{\nu_N}[F_N \mid \boldsymbol{\eta}_{S\setminus\{x,y\}} = \boldsymbol{\sigma},\ |\boldsymbol{\eta}(x)| = p].$$

- The hypergeometric weights combine the two species increments into

$$\overline{F}_N^{\boldsymbol{\sigma},p+1} - \overline{F}_N^{\boldsymbol{\sigma},p}.$$

- Telescoping

$$\sum_{p:\text{near } \mathcal{D}_N^y}^{\text{near } \mathcal{D}_N^x} (\overline{F}_N^{\boldsymbol{\sigma},p+1} - \overline{F}_N^{\boldsymbol{\sigma},p}) = \overline{F}_N^{\boldsymbol{\sigma},\text{near } \mathcal{D}_N^x} - \overline{F}_N^{\boldsymbol{\sigma},\text{near } \mathcal{D}_N^y}.$$

Conditional averaging

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- Telescoping

$$\sum_{p:\text{near } \mathcal{D}_N^y}^{\text{near } \mathcal{D}_N^x} (\overline{F}_N^{\boldsymbol{\sigma},p+1} - \overline{F}_N^{\boldsymbol{\sigma},p}) = f_N(x) - f_N(y) + o(1)$$

  and we can extend the estimate to the wells $\mathcal{E}_N^x$ and $\mathcal{E}_N^y$.

$\mathfrak{R}^{(2)}$ completion

Summing all contributions,

$$\begin{aligned}\theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} &= \nu_N(\mathcal{E}_N^x) \sum_{y \neq x} \mathfrak{r}(x,y)[f_N(x) - f_N(y)] + o(1) \\ &= -\nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x) + o(1).\end{aligned}$$

Therefore, resolvent equation becomes

$$\lambda f_N(x) \nu_N(\mathcal{E}_N^x) - \nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x) = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x) + o(1).$$

Since $\nu_N(\mathcal{E}_N^x) \to |S_\star|^{-1}$,

$$(\lambda - \mathfrak{L}_{\mathbb{X}}) f_N(x) = \mathbf{g}(x) + o(1), \quad \forall x \in S_\star.$$

The $\mathfrak{R}^{(2)}$ condition

Let $(\lambda - \mathfrak{L}_{\mathbb{X}})\mathfrak{f} = \mathbf{g}$, then we obtain

$$\|f_N - \mathfrak{f}\|_\infty \xrightarrow[N \to \infty,\ \epsilon \downarrow 0]{} 0.$$

Further topics

Q. Generalize for super-critical k-species case?

With time scale $\theta_N = N^{\alpha+1}$,

$$\mathfrak{r}(x,y) = \frac{1}{\Gamma(\alpha)\mathcal{I}_\alpha} \left(\sum_{j=1}^{k} \frac{\rho_j}{\mathrm{cap}_j(x,y)} \right)^{-1}$$

Q. Can we obtain analogous results at $\alpha = 1$?

The difficult part is the local metastability for larger wells.

Thank you!